By 1939, when Oppenheimer and Snyder discovered these things, physicists had become accustomed to the fact that time is relative; the flow of time is different as measured in different reference frames that move in different ways through the Universe. But never before had anyone encountered such an extreme difference between reference frames. That *the implosion freezes forever as measured in the static, external frame but continues rapidly on past the freezing point as measured in the frame of the star's surface* was extremely hard to comprehend. Nobody who studied Oppenheimer and Snyder's mathematics felt comfortable with such an extreme warpage of time. Yet there it was in their formulas. One might wave one's arms with heuristic explanations, but no explanation seemed very satisfying. It would not be fully understood until the late 1950s (near the end of this chapter).

By looking at Oppenheimer and Snyder's formulas from the viewpoint of an observer on the star's surface, one can deduce the details of the implosion even after the star sinks within its critical circumference; that is, one can discover that the star gets crunched to infinite density and zero volume, and one can deduce the details of the spacetime curvature at the crunch. However, in their article describing their calculations, Oppenheimer and Snyder avoided any discussion of the crunch whatsoever. Presumably Oppenheimer was prevented from discussing it by his own innate scientific conservatism, his unwillingness to speculate (see the last two paragraphs of Chapter 5).

If reading the star's final crunch off their formulas was too much for Oppenheimer and Snyder to face, even the details outside and at the critical circumference were too bizarre for most physicists in 1939. At Caltech, for example, Tolman was a believer; after all, the predictions were unequivocal consequences of general relativity. But nobody else at Caltech was very convinced. General relativity had been tested experimentally only in the solar system, where gravity is so weak that Newton's laws give almost the same predictions as general relativity. By contrast, the bizarre Oppenheimer–Snyder predictions relied on ultra-strong gravity. General relativity might well fail before gravity ever became so strong, most physicists thought; and even if it did not fail, Oppenheimer and Snyder might be misinterpreting what their mathematics was trying to say; and even if they were not misinterpreting their mathematics, their calculation